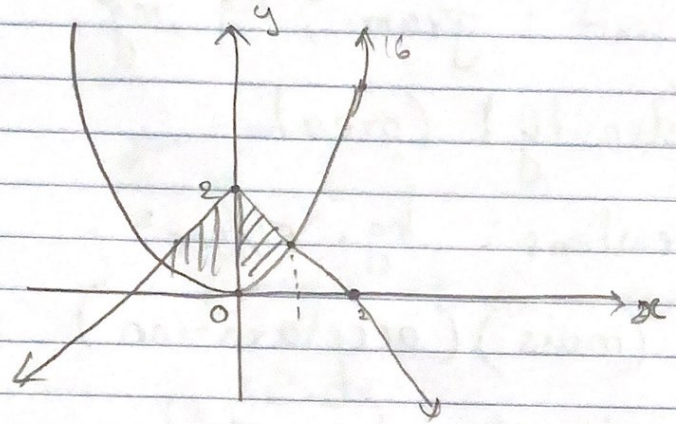
Quiz :-

C1 :

$y = x^4$ and $y = 2 - |x|$



$x^4 = 2 - |x| \Rightarrow x^4 + x - 2 = 0$

$x(x^3+1) - 2 = 0 \Rightarrow x(x^3+1) = 2$, $x = 2$

$x^3 + 1 = 2 \Rightarrow x^3 = 1 \Rightarrow x = \sqrt[3]{1} = x = 1$

$$A = \int_{-1}^{1} \left[2 - (x) - x^4 \right] dx$$

$$A = \int_{-1}^{1} (2 - x - x^4) \, dx$$

$$A = \left[2x - \frac{x^2}{2} - \frac{x^5}{5} \right]_{-1}^{1}$$

$$A = \left[2 - \frac{1}{2} - \frac{1}{5} \right] - \left[-2 + \frac{1}{2} + \frac{1}{5} \right]$$

$$A = 2\left(\frac{3}{2} - \frac{1}{5} \right) \Rightarrow A = 2\left(\frac{15 - 2}{10} \right)$$

$$A = \pi\left(\frac{13}{10}\right)$$

$$A = \frac{13}{5}$$